

Mohamad Mahdi Mehralian

Full-Stack Engineer / Frontend Engineer (React · Next.js)

Portfolio: github.com/mohamadmahdiMN/portfolio

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PROFILE

Computer Engineering senior and full-stack TypeScript engineer who ships production-grade systems end to end. Built a **real-time collaborative IDE** with CRDT-based conflict-free sync, a **queue-driven sandbox** that executes untrusted code inside hardened Docker containers, and a **monorepo e-commerce platform** on Next.js, NestJS/Express, PostgreSQL and Prisma. Strong in distributed real-time architecture, secure API design, and container hardening.

TECHNICAL SKILLS

LANGUAGES	TypeScript · JavaScript (ES6+) · SQL · HTML5 · CSS3
FRONTEND	Next.js (App Router) · React · Tailwind CSS · Monaco Editor · State Management · Responsive Web Design
BACKEND	Node.js · NestJS · Express.js · RESTful APIs · WebSockets (Socket.io) · JWT Auth
DATA	PostgreSQL · Prisma ORM · Supabase · Redis (Pub/Sub)
SYSTEMS	Docker · Container Hardening · BullMQ · Yjs (CRDTs) · Git · GitHub · Vercel

FEATURED PROJECTS

DevPair -- Real-Time Collaborative IDE & Execution Sandbox

Next.js · NestJS · TypeScript · Yjs (CRDTs) · Socket.io · BullMQ · Redis · Docker · PostgreSQL · Prisma
[mohamadmahdiMN/DevPair](https://github.com/mohamadmahdiMN/DevPair)

- **Architected** a real-time collaborative code editor using Yjs CRDTs over WebSockets, enabling concurrent multi-user typing with conflict-free state convergence and zero central locking.
- **Engineered** an asynchronous execution pipeline on BullMQ and Redis that runs untrusted user code inside isolated Docker containers hardened with a disabled network namespace, enforced memory/CPU ceilings, and a 5-second execution timeout.
- **Configured** a Redis Pub/Sub adapter for NestJS WebSocket gateways, allowing real-time socket sessions to scale horizontally across multiple stateless server instances.

Modern Marketplace -- Monorepo E-Commerce Platform

Next.js · Express.js · TypeScript · Prisma ORM · PostgreSQL · Tailwind CSS · JWT
[mohamadmahdiMN/marketplace](https://github.com/mohamadmahdiMN/marketplace)

- **Designed** and deployed a full-stack e-commerce monorepo cleanly separating Next.js frontend services from a modular Express REST API backend.
- **Implemented** secure JWT authentication with bcrypt password hashing, dynamic category mapping, and multipart file upload handlers for listing media.
- **Leveraged** Prisma ORM for type-safe relational querying and authored idempotent seeding scripts to provision reproducible test listings and user schemas.

Accountability AI -- Automated Progress-Tracking SaaS

Next.js (App Router) · Supabase · TypeScript · Resend API · Vercel Cron
[mohamadmahdiMN/accountability-ai](https://github.com/mohamadmahdiMN/accountability-ai)

- **Developed** an automated email accountability SaaS on the Next.js App Router, with Supabase handling authentication and relational data persistence.
- **Integrated** serverless CRON jobs with the Resend API to dispatch scheduled daily progress triggers and personalized user goal reports.

Assembly Endgame -- Interactive Game Application

React · TypeScript · Tailwind CSS
[mohamadmahdiMN/assembly-endgame](https://github.com/mohamadmahdiMN/assembly-endgame)

- **Built** an interactive word-guessing game using modern React state management, custom hooks for conditional rendering logic, and a fully responsive UI.

EDUCATION

B.S. in Computer Engineering

Expected 2026

Islamic Azad University, Najafabad Branch · Focus: distributed systems, data structures & algorithms, database design, software architecture

Languages: English (C1 Advanced) · Persian (Native)